

$$(uv)^n = {}^nC_0 u^0 v^n + {}^nC_1 u^1 v^{n-1} + {}^nC_2 u^2 v^{n-2} + {}^nC_3 u^3 v^{n-3}$$

$$\text{also} = {}^nC_0 u^1 v^n + {}^nC_1 u^2 v^{n-1} + {}^nC_2 u^3 v^{n-2} + {}^nC_3 u^4 v^{n-3}$$

Q) find $\frac{d^3 y}{dx^3}$ of $n^4 \cdot \sin x$ $\therefore x$ is always a single variable

$$u^0 = \sin x$$

$$v^0 = x^4$$

$$u^1 = \cos x$$

$$v^1 = 4x^3$$

$$u^2 = -\sin x$$

$$v^2 = 12x^2$$

$$u^3 = -\cos x$$

$$v^3 = 24x$$

$$u^4 = \sin x$$

$$v^4 = 24$$

$${}^nC_0 u^1 v^0 + {}^nC_1 u^2 v^1 + {}^nC_2 u^3 v^2 + {}^nC_3 u^4 v^3 + {}^nC_4 u^0 v^4$$

$$= (\sin x)(x^4) + 4(-\cos x)(4x^3) + 6(-\sin x)(12x^2) + 4(-\cos x)(24x) + (1)(24)(\sin x)$$

$$= x^4 \sin x - 16x^3 \cos x - 72x^2 \sin x + 96x \cos x + 24 \sin x$$

Q) find $\frac{d^3 y}{dx^3}$ of $3x^2 e^x + 2x e^x$

$$= 3x^2 e^x + 2x e^x$$

$$= \underset{u}{e^x} (\underset{v}{3x^2 + 2x})$$

$$u^0 = e^x$$

$$v^0 = 3x^2 + 2x$$

$$u^1 = e^x$$

$$v^1 = 6x + 2$$

$$u^2 = e^x$$

$$v^2 = 6$$

$${}^nC_0 u^3 v^0 + {}^nC_1 u^2 v^1 + {}^nC_2 u^1 v^2$$

$$= 1(e^x)(3x^2 + 2x) + 2(e^x)(6x + 2) + 1(e^x)(6)$$

$$= e^x (3x^2 + 2x) + 2e^x (6x + 2) + e^x (6)$$

$$= e^x (3x^2 + 2x + 12x + 4 + 6)$$

$$= e^x (3x^2 + 14x + 10)$$

Q) find $\frac{dy}{dx}$ for $\frac{e^{2x} \sin x}{u \cdot v}$

$$u^0 = e^{2x}$$

$$v^0 = \sin x$$

$$u^1 = 2e^{2x}$$

$$v^1 = \cos x$$

$$u^2 = 4e^{2x}$$

$$v^2 = -\sin x$$

$$u^3 = 8e^{2x}$$

$$v^3 = -\cos x$$

$$u^4 = 16e^{2x}$$

$$v^4 = \sin x$$

$$u^5 = 32e^{2x}$$

$$v^5 = \cos x$$

$$u^6 = 64e^{2x}$$

$$v^6 = -\sin x$$

$$u^7 = 128e^{2x}$$

$$v^7 = -\cos x$$

$$= {}^nC_0 u^0 v^7 + {}^nC_1 u^1 v^6 + {}^nC_2 u^2 v^5 + {}^nC_3 u^3 v^4 + {}^nC_4 u^4 v^3 + {}^nC_5 u^5 v^2 + {}^nC_6 u^6 v^1 + {}^nC_7 u^7 v^0$$

$$= (1)(128e^{2x})(\sin x) + (7)(64e^{2x})(\cos x) + (21)(32e^{2x})(-\sin x) + (35)(16e^{2x})(-\cos x) + (35)($$

Accidentally did this for 7th derivative.

so I give up now.